

# MATHEMATICS

TARGET : JEE- Advanced

# CAPS-18

AREA UNDER CURVE

## SCQ (Single Correct Type) :

- Let  $f(x) = 2\sqrt{x}$  and  $g(x) = 2\sqrt{1-x}$  be two functions then the area bounded by  $y = f(x)$ ,  $y = g(x)$  and  $x$  axis is  
 (A)  $\frac{1}{3\sqrt{2}}$  sq unit (B)  $\frac{2}{3\sqrt{2}}$  sq unit (C)  $\frac{1}{\sqrt{2}}$  sq unit (D)  $\frac{4}{3\sqrt{2}}$  sq unit
- The area bounded by the lines  $y = 2$ ,  $x = 1$ ,  $x = a$  and the curve  $y = f(x)$ , which cuts the last two lines above the first line for all  $a \geq 1$ , is equal to  $\frac{2}{3}((2a)^{3/2} - 3a + 3 - 2\sqrt{2})$  then  $f(x) =$   
 (A)  $2\sqrt{2x}$ ;  $x \geq 1$  (B)  $\sqrt{2x}$ ;  $x \geq 1$  (C)  $2\sqrt{x}$ ;  $x \geq 1$  (D) None of these
- A variable circle  $C$  touches  $y = 0$  and passes through the point  $(0,1)$ . Let the locus of the centre of  $C$  is the curve  $P$ . The area enclosed by the curve  $P$  and the line  $x + y = 2$  is  
 (A)  $\frac{14}{3}$  (B)  $\frac{16}{3}$  (C)  $\frac{15}{3}$  (D)  $\frac{11}{3}$
- Let  $n S_n$  be the area of the figure enclosed by a curve  $y = x^3(1-x^2)^n$ ,  $0 \leq x \leq 1$  and the  $x$ -axis. The value of  $\lim_{n \rightarrow \infty} \sum_{k=1}^n S_k$  is  
 (A)  $\frac{1}{2}$  (B)  $\frac{1}{3}$  (C)  $\frac{1}{4}$  (D) 1
- Find the area of the region enclosed between the curve  $y = 2x^4 - x^2$ , the  $x$ -axis and the ordinates of the points where the curve has local minima  
 (A)  $\frac{7}{60}$  (B)  $\frac{7}{80}$  (C)  $\frac{7}{120}$  (D)  $\frac{7}{160}$
- Let  $f(x) = \begin{cases} \ln(x - [x]); & x \notin I \\ 0; & x \in I \end{cases}$  Area enclosed by curves  $y = f(x)$ ,  $y = f(|x|)$ ,  $-5 \leq x \leq 5$  and  $y = 0$  is  
 (A)  $5(1 - \ln 2)$  (B) 0 (C)  $5 \ln 2$  (D) none of these
- The area bounded by the curves  $y = 2 - |x - 1|$ ,  $y = \sin x$ ;  $x = 0$  and  $x = 2$  is \_\_\_\_\_.  
 (A)  $1 + 2\cos^2 1$  (B)  $2 + \sin^2 1$  (C)  $\frac{\pi}{2}$  (D)  $1 + \log 2$
- The area bound by the curve  $y = f(x)$ , the  $x$ -axis and the line  $y = 1$ , where  $f(x) = 1 + \frac{1}{x} \int_1^x f(t) dt$ , ( $x \neq 0$ ) is \_\_\_\_\_.  
 (A)  $2\left(1 - \frac{1}{e}\right)$  (B)  $2\left(1 + \frac{1}{e}\right)$  (C)  $\left(1 + \frac{1}{e}\right)$  (D)  $\left(1 - \frac{1}{e}\right)$



9. Given  $f(x) = x^3 + x - 2$ ,  $f: \mathbb{R} \rightarrow \mathbb{R}$ , area of region bounded by  $y = f(x)$ ,  $y = f^{-1}(x)$  and  $xy = 0$  in 1<sup>st</sup> quadrant is A, where  $A = 3t - \frac{5}{2}$ , where  $t = a^{1/b}$  and  $3(a+b) = n$ ;  $a, b, n \in \mathbb{N}$  and  $a$  is prime. Difference of last two digits of  $a^{a^{11}-36}$  is
- (A) 0 (B) 1 (C) 2 (D) 3

**MCQ (One or more than one correct) :**

10. The area bounded by the curve  $y = f^{-1}(x)$ ,  $x$ -axis,  $x = 0$  and  $x = \pi$ , where  $f(x)$  is given by  $f(x) = x + \sin x$  is  $\frac{\pi^2 - a}{b}$  then the correct statement are :
- (A)  $a = 2b$  (B)  $a + 2b = 8$  (C)  $3a = b$  (D)  $\left(\frac{2\pi}{b}, \frac{a\pi}{4}\right)$  lies on  $f(x)$
11. If  $A_i$  is the area bounded by  $|x - a_i| + |y| = b_i$ ,  $i \in \mathbb{N}$  where  $1 \leq a_{i+1} = a_i + \frac{3}{2}b_i$  and  $b_{i+1} = \frac{b_i}{2}$ ,  $a_1 = 0$ ,  $b_1 = 32$ , then \_\_\_\_.
- (A)  $A_3 = 128$  (B)  $A_3 = 256$  (C)  $\lim_{n \rightarrow \infty} \sum_{i=1}^n A_i = \frac{8}{3}(32)^2$  (D)  $\lim_{n \rightarrow \infty} \sum_{i=1}^n A_i = \frac{4}{3}(16)^2$
12. Let  $T$  be the triangle with vertices  $(0, 0)$ ,  $(0, c^2)$  and  $(c, c^2)$  and let  $R$  be the region between  $y = cx$  and  $y = x^2$  where  $c > 0$  then
- (A)  $\text{Area}(R) = \frac{c^3}{6}$  (B)  $\text{Area of } R = \frac{c^3}{3}$  (C)  $\lim_{c \rightarrow 0^+} \frac{\text{Area}(T)}{\text{Area}(R)} = 3$  (D)  $\lim_{c \rightarrow 0^+} \frac{\text{Area}(T)}{\text{Area}(R)} = \frac{3}{2}$

**Comprehension Type Question:**

**Comprehension # 1**

Let  $f: \mathbb{R} \rightarrow \mathbb{R}$  be continuous function and bijective, defined such that  $f(\alpha) = 0$  ( $\alpha \neq 0$ ). The area bounded by  $y = f(x)$ ,  $x = \alpha$ ,  $x = \alpha - t$  is equal to area bounded by  $y = f(x)$ ,  $x = \alpha$ ,  $x = \alpha + t$ ,  $\forall t \in \mathbb{R}$  then

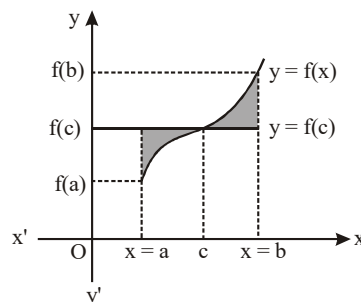
13.  $y = f(x)$  is symmetrical about the point
- (A)  $(0, 0)$  (B)  $(0, \alpha)$  (C)  $(\alpha, 0)$  (D)  $(\alpha, \alpha)$
14.  $f(2\alpha)$  equal to \_\_\_\_.
- (A)  $f(\alpha)$  (B)  $-f(\alpha)$  (C)  $f(0)$  (D)  $-f(0)$

**Comprehension # 2**

If  $y = f(x)$  is a monotonic function in  $(a, b)$ , then the area bounded by the ordinates at  $x = a$ ,  $x = b$ ,  $y = f(x)$  and  $y = f(c)$  (where  $c \in (a, b)$ ) is minimum when  $c = \frac{a+b}{2}$ .

**Proof :**  $A = \int_a^c (f(c) - f(x)) dx + \int_c^b (f(x) - f(c)) dx$

$$= f(c)(c-a) - \int_a^c f(x) dx + \int_c^b f(x) dx - f(c)(b-c)$$



$$\Rightarrow A = [2c - (a + b) f(c) + \int_c^b (f(x)) dx - \int_a^c (f(x)) dx]$$

Differentiating w.r.t. c,

$$\frac{dA}{dc} = [2c - (a + b)] f'(c) + 2f(c) + 0 - f(c) - (f(c))$$

for maxima and minima  $\frac{dA}{dc} = 0$

$$\Rightarrow f'(c)[2c - (a + b)] = 0 \text{ (as } f'(c) \neq 0 \text{)}$$

$$\text{hence } c = \frac{a + b}{2}$$

$$\text{Also } c < \frac{a + b}{2}, \frac{dA}{dc} < 0 \text{ and } c > \frac{a + b}{2}, \frac{dA}{dc} > 0.$$

Hence A is minimum when  $c = \frac{a + b}{2}$ .

15. If the area bounded by  $f(x) =$  and the straight lines  $x = 0$ ,  $x = 2$  and the x-axis is minimum, then the value of  $a$  is  
 (A)  $\frac{1}{3}$  (B) 2 (C) 1 (D)  $\frac{2}{3}$
16. The value of the parameter  $a$  for which the area of the figure bounded by the abscissa axis, the graph of the function  $y = x^3 + 3x^2 + x + a$  and the straight lines, which are parallel to the axis of ordinates and cut the abscissa axis at the point of extremum of the function, is the least, is  
 (A) 2 (B) 0 (C) -1 (D) 1
17. If the area enclosed by  $f(x) = \sin x + \cos x$ ,  $y = a$  between the two consecutive points of extremum is minimum, then the value of  $a$  is  
 (A) 0 (B) -1 (C) 1 (D) 2

### Comprehension # 3

Consider the areas  $S_0, S_1, S_2, \dots$  bounded by the x-axis and half-waves of the curve  $y = e^{-x} \sin x$ , where  $x \geq 0$ .

18. The value of  $S_0$  is  
 (A)  $\frac{1}{2}(1 + e^\pi)$  sq. units (B)  $\frac{1}{2}(1 + e^{-\pi})$  sq. units  
 (C)  $\frac{1}{2}(1 - e^{-\pi})$  sq. units (D)  $\frac{1}{2}(e^\pi - 1)$  sq. units.
19. The sequence  $S_0, S_1, S_2, \dots$ , forms a G.P. with common ratio  
 (A)  $\frac{e^\pi}{2}$  (B)  $e^{-\pi}$  (C)  $e^\pi$  (D)  $\frac{e^{-\pi}}{2}$
20.  $\sum_{n=0}^{\infty} S_n$  is equal to  
 (A)  $\frac{1 + e^\pi}{1 - e^{-\pi}}$  (B)  $\frac{1}{2}(1 + e^\pi)$  (C)  $\frac{1}{2(1 - e^{-\pi})}$  (D) None of these

**Numerical based Questions :**

21. If A is the area of the figure bounded by the straight line  $x = 0$  and  $x = 2$  and the curves  $y = 2^x$  and  $y = 2x - x^2$ , then the value of  $6\left(\frac{3}{\ln 2} - A\right)$  is
22. The graph of  $y^2 + 2xy + 40|x| = 400$  divides the plane into regions. The area of the bounded region is
23. Area enclosed by the curve  $4 \leq x^2 + y^2 \leq 2(|x| + |y|)$  is (in sq. units)
24. The area bounded by the curves  $y = -\sqrt{4 - x^2}$ ,  $x^2 = -y\sqrt{2}$  and  $x = y$  can be expressed in the form of  $\pi + \frac{\lambda}{\mu}$ , ( $\lambda, \mu$  being relatively prime positive integers) then  $\lambda + \mu$  equals to \_\_\_\_\_.
25. Find the area enclosed between the curves :  $y = \log_e(x + e)$ ,  $x = \log_e(1/y)$  & the x-axis.

**Matrix Match Type :**

26. Match the following

| Column-I                                                                                                                                                                                                                                             | Column-II |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| (a) Let $f(K) = \frac{K}{2009}$ and $g(K) = \frac{(f(K))^4}{(1 - f(K))^4 + (f(K))^4}$ and $S = \sum_{K=0}^{2009} g(K)$ then sum of the diagonal in S =                                                                                               | (p) 0     |
| (b) $f : D \rightarrow \mathbb{R}$ such that $f(k) = \sqrt{\sin(\cos x)} + \log_e(-2\cos^2 x + 3\cos x + 1)$ then $\int_{x_1}^{x_2} \left[\cos x - \frac{1}{2}\right] dx$ is equal to, $x_1, x_2 \in D$ and $[.]$ denotes greatest integer function. | (q) 1     |
| (c) The area of region, represented by $ x  +  y  \leq 2$ and $x^2 \leq y$ is S square units then $3S =$ _____.                                                                                                                                      | (r) 7     |
| (d) $\lim_{y \rightarrow \infty} \frac{\int_1^y [\tan^{-1} x] dx}{\int_1^y \left[1 + \frac{1}{x}\right] dx} =$ _____. ( $[.]$ denotes greatest integer function)                                                                                     | (s) 6     |

(A)  $A \rightarrow s$ ;  $B \rightarrow p$ ;  $C \rightarrow s$ ;  $D \rightarrow q$

(B)  $A \rightarrow p$ ;  $B \rightarrow s$ ;  $C \rightarrow r$ ;  $D \rightarrow q$

(C)  $A \rightarrow s$ ;  $B \rightarrow p$ ;  $C \rightarrow r$ ;  $D \rightarrow q$

(D)  $A \rightarrow s$ ;  $B \rightarrow p$ ;  $C \rightarrow r$ ;  $D \rightarrow s$

27. Column - I

Column-II

(A) The area bounded by the curve  $y = x|x|$ , x-axis  
and the ordinates  $x = 1$ ,  $x = -1$

(P)  $\frac{10}{3}$  sq. units

(B) The area of the region lying between the line  $x - y + 2 = 0$   
and the curve  $x = \sqrt{y}$

(Q)  $\frac{64}{3}$  sq. units

(C) The area enclosed between the curves  $y^2 = x$  and  $y = |x|$

(R)  $\frac{2}{3}$  sq. units

(D) The area bounded by parabola  $y^2 = x$ ,  
straight line  $y = 4$  and y-axis

(S)  $\frac{1}{6}$  sq. units

(A) A→Q; B→S; C→P; D→R

(B) A→S; B→R; C→Q; D→P

(C) A→R; B→P; C→S; D→Q

(D) A→Q; B→R; C→P; D→S

ANSWER KEY OF CAPS-18

|          |           |         |         |           |
|----------|-----------|---------|---------|-----------|
| 1. (D)   | 2. (A)    | 3. (B)  | 4. (C)  | 5. (C)    |
| 6. (A)   | 7. (A)    | 8. (A)  | 9. (D)  | 10. (ABD) |
| 11. (AC) | 12. (AC)  | 13. (C) | 14. (D) | 15. (D)   |
| 16. (C)  | 17. (A)   | 18. (A) | 19. (B) | 20. (B)   |
| 21. (8)  | 22. (800) | 23. (8) | 24. (4) | 25. (2)   |
| 26. (C)  | 27. (C)   |         |         |           |